1.7 Inverse Trigonometric Functions

Name _____ Period ____ Date _____

Pre-Calculus 1.7 · Student sheet

I can…

- evaluate inverse trigonometric functions exactly and with a calculator.
- use the inverse properties of trigonometric functions.
- evaluate compositions of trigonometric and inverse trigonometric functions.

Vocabulary · fill in the missing word

Inverse sine function — y = _____ x means sin y = x, with $-\pi/2 \le y \le \pi/2$

Inverse cosine function — y = _____ x means cos y = x, with $0 \le y \le \pi$

Inverse tangent function — y = _____ x means tan y = x, with $-\pi/2 < y < \pi/2$

Part 1 · Restrict the domain

Graph y = sin x and y = 1/2.

Question	Answer
How many x-values give sin x = 1/2?	
Which one is in $[-\pi/2, \pi/2]$?	
An interval where cos x passes the Horizontal Line Test	

Part 2 · Notes and examples (fill in each note, then work the example)

arcsin x gives the angle in $[-\pi/2, \pi/2]$ whose sine is x. **arccos x** gives the angle in _____.

Example 1

Evaluate.

a. $\arcsin\left(-\frac{1}{2}\right)$

b. $\arccos(-1)$

c. $\arctan 1$

d. $\arcsin 2$

Example 2

Use a calculator in radian mode.

a. $\arctan(-8.45)$

b. $\arcsin 0.2447$

Inverse properties: $\sin(\arcsin x) = x$ for $-1 \le x \le 1$; $\arcsin(\sin y) = y$ only for y in _____.

Example 3

Evaluate, if possible.

a. $\tan(\arctan(-5))$

b. $\arcsin(\sin\frac{5\pi}{3})$

c. $\cos(\cos^{-1}\pi)$

Compositions: draw a right triangle for the inside angle, then use _____.

Example 4

Find the exact value.

a. $\tan(\arccos\frac{2}{3})$

b. $\cos(\arcsin(-\frac{3}{5}))$

Part 3 · Partner practice

Partner 1

Evaluate $\arccos(\frac{\sqrt{3}}{2})$ and $\arctan(-\sqrt{3})$.

Partner 2

Find $\sin(\arctan\frac{3}{4})$.